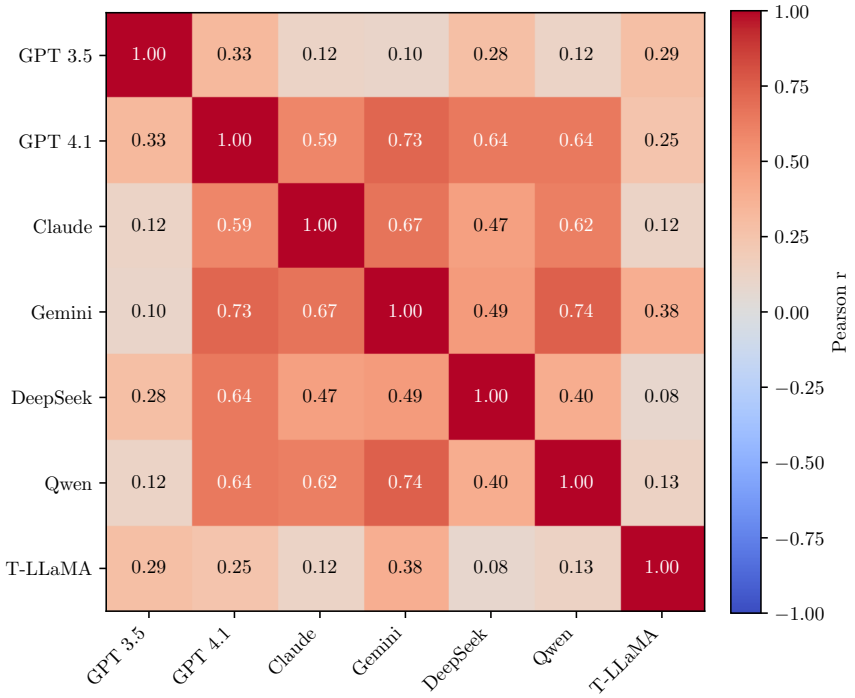

Figure 15: Pearson correlation matrix between models' tool-selection bias patterns.

G.4 Table related to the Reduction of Bias due to Mitigation

Table 5 demonstrates that our mitigation substantially flattens selection distributions and reduces both API- and position-level bias.

Table 5: Average cluster-level API bias δ_{API}, positional bias δ_{pos}, and combined bias δ_{model} before and after mitigation.

Setup	δ_{API}	δ_{pos}	δ_{model}
Before	0.338	0.422	0.380
After	0.108	0.079	0.094

H Effect of Metadata Perturbation on Bias

Relative to the base distributions, both models move farther from uniform (get more biased) when we lower semantic signal (see bars corresponding to the description + parameter and full perturbations in Figure 18). For Gemini, these manipulations yield the largest TV distances to uniform (≈0.42–0.43); ChatGPT shows similar results. This suggests that when descriptions/parameters are corrupted, models amplify bias rather than flatten choices.

Conversely, targeted edits to the most popular API tends to decrease bias. For Gemini, targeting the description of the most popular API leads to an average TVD slightly below the baseline and swapping the description between most- and least popular API leads to one that is substantially lower, indicating that weakening or transferring the strongest semantic cue moves the selection

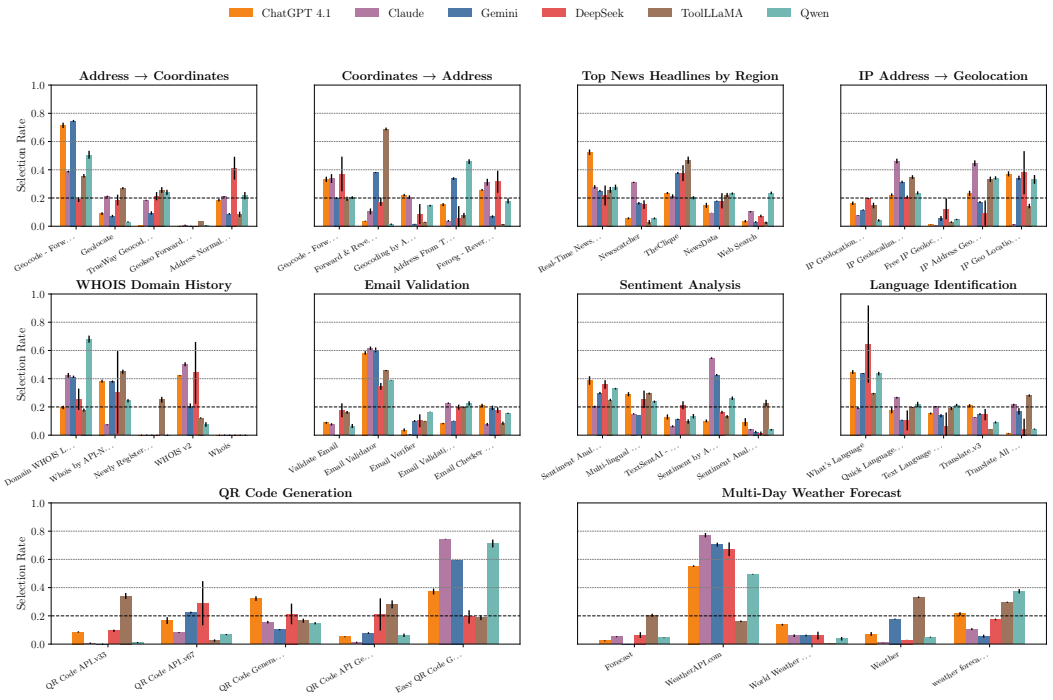

Figure 16: Selection distributions for six LLMs across ten clusters of functionally equivalent APIs. Each subplot corresponds to one cluster, with the x-axis indicating the API in the cluster and the y-axis showing the (mean) fraction of times each model chose that API over 500 inference runs; error bars indicate the standard deviation across three independent experimental runs. This visualization highlights how different models exhibit systematic preferences for some APIs.

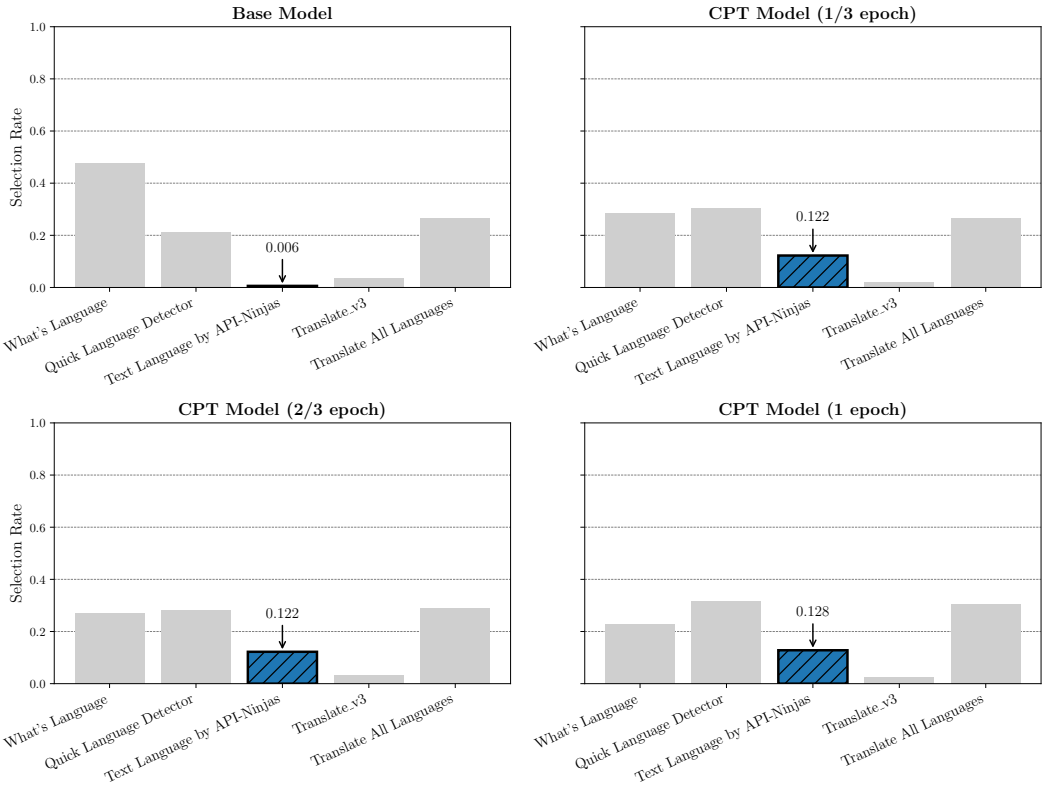

Figure 17: Selection rates for the Language Identification cluster across biased continued pre-training (CPT) checkpoints. Bars give the fraction that each endpoint was chosen across 500 inference runs. Panels show (top-left) base model, (top-right) CPT after 1/3 epoch, (bottom-left) 2/3 epoch, and (bottom-right) 1 full epoch. The Text Language by API-Ninjas endpoint is highlighted, with its exact selection rate printed above its bar. Differences across panels visualize how biased CPT shifts tool choice over training.

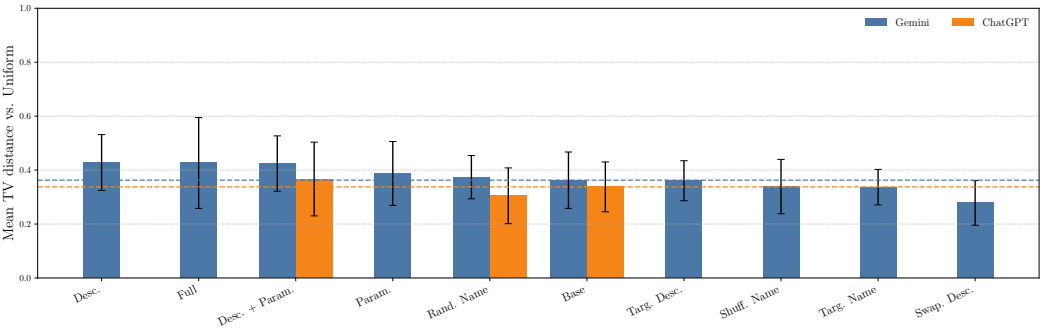

Figure 18: Mean total-variation (TV) distance from the uniform selection distribution to the distribution pertaining to each metadata perturbation (higher = more deviation from uniform). Blue bars show Gemini and orange bars show ChatGPT; error bars denote standard deviation across clusters, and single bars indicate perturbations not run for ChatGPT. Dashed horizontal lines (in the corresponding model colors) mark each model's baseline TV-to-uniform without perturbations.

distribution toward uniform. Name-only manipulations have similar effects, but name scrambling does not increase bias as much.